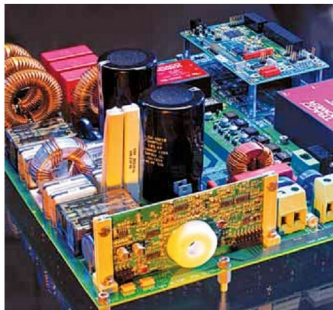


The Power of Power Electronics

Demand for electrical and electronics engineers in the field of power electronics is steadily on the rise in India. But what is required of them to befit this industry? Let's look at the current scenario, including entry-level roles, typical pay-packages, skills expected and much more



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What is the one thing that the Indian power sector wishes to eradicate? Power cuts, of course. This is the most common, aggravating and discomforting problem faced by hundreds of millions of people across India.

As a result, "One of the current focus areas in the power sector is driving energy conservation at large power plants, where auxiliary power consumes close to 10 per cent of what is generated," says Prakash Nayak, chairman, IET India Power Engineers Panel.

"Many countries including India, of late, have been deploying variable-frequency and highly energy-efficient drives to improve the energy efficiency of the motors or pumps which are energy guzzlers," Nayak adds.

On the renewable side of energy,

India is fortunate to have over 3000 hours of sunshine annually. Nayak explains, "Solar energy can be utilised as an alternative source of energy, with the help of photovoltaic or concentrated solar thermal power technology. This technology uses extensive power electronics devices and systems; one example is transformerless high-efficiency inverters ranging

from small capacity such as 300W to hundreds of megawatts."

Talking about yet another renewable source of energy—wind—Nayak shares "India has a large potential of the order of 40 GW per year. The cost, initially, might be on higher side, however, it is coming down at a much faster rate and will reach grid parity by 2017."

Commenting on the evolution of power electronics, he says, "In order to have efficient bulk transmission from generating hub to industrial hub, these hubs deploy IGBT-based HVDC converters at each of the substations, static VAR compensations, etc."

Drives for increased energy efficiency in power plants, inverters used in both wind and solar power generation and converters used in HVDC transmission for bulk transfer of power are all based on power electronics technology.

Another emerging area is DC grid, which deploys devices like solidstate transformers and switchgears.

Career opportunities in this field

"In addition to the industrial sector, power electronics is rapidly gaining momentum in automotive space too with the advent of electric and hybrid vehicles," says Basavaraj Garadi, chief expert, Robert Bosch Engineering and Business Solutions.

Engineers interested in this field are required to have experience with converters, rectifiers and inverters (AC and DC) and should be experts in specific fields across the engineering life-cycle, including design, test and support. Specialist areas could include switch-mode power supply, motor control, IGBT/MOSFET design and battery charging system.

"DC grid opens tremendous opportunities for power electronics engineers and researchers. Activities related to power semiconductor development, product development, and also system simulation and power system development are surely going to create greater career opportunities in the field of power electronics," informs Nayak.

Sharing his views on why one should opt for this field of engineering, Garadi explains, "In the past, there was none or minimal direct involvement of electronics in power generation. With the advent of solar energy, power electronics is the key element in PV plants." Motor control is another system that is widely applied in industrial and automotive applications.

"Three-phase induction motors and BLDC motors are hot topics," says Garadi. Variable-frequency drives find